

Reaffirming the Critical Role of Transformative Research and Knowledge Production in the Age of Post-Truth



Seismograph Motion Detector Alarm For Detecting Seismic Waves and Mitigating Earthquake Hazards Using Arduino Uno (MDASH-UNO)

Abstract: The Philippines' location on the Pacific Ring of Fire makes it vulnerable to natural disasters like earthquakes. An earthquake is an abrupt movement of rock materials beneath the earth's surface that causes a weak to violent shaking of the ground. To identify, amplify, and record both ground motion and earthquakes, a seismograph is utilized. This study aims to create a device that will alert people in the event of a weak to violent shaking of the ground. Using the Arduino Uno Board as the central component of the device, the ADXL335 3 axis accelerometer detects all motion in 3 dimensions; wires are used for connections; an LED and buzzer act as the outgoing indicators; and an LCD display, displays the accelerometer's detect motion. Conceptualization and visualization, requirement analysis, planning, quick design, programming, and testing are the stages that MDASH-UNO must go through. The MDASH-UNO device underwent vibration (SV) testing by the Department of Science and Technology (DOST) MIRDC – ATL. It underwent tests to evaluate whether or not it could detect motion using a motion sensor to determine the frequency of a motion and send an alert through the buzzer's sound emission and led blinking to alert people about the motion being detected. As it did so, it was also able to convey an alarm through the buzzer's sound emission which could help reduce the risk of earthquake hazard by alerting people about the motion detected by the sensor

Keywords: seismograph; Arduino UNO; earthquake; sine vibration; IoT devices

1. INTRODUCTION

When pressures from the tectonic driving forces of plate motion and mountain building build up within the Earth's crust, they suddenly release, causing earthquakes. Two main categories of quantifiable physical phenomena are produced by earthquakes. Initially, seismic waves spread outward from the epicenter of the earthquake. Tilloy et al. (2019) stated that, as a result of its location on the Pacific Ring of Fire, the Philippines is vulnerable to natural calamities, including earthquakes and volcanic eruptions. Earthquake: An abrupt movement of rock materials beneath the surface of the earth that causes a weak to strong shaking of the ground. There are risks associated with earthquakes, including (1) ground rupture, which is the movement of the ground caused by a fault; (2) ground shaking, which is the violent up-down and sideways motion experienced during an earthquake; (3) liquefaction, which is the behavior of body sediments turning into liquid as a result of extremely intense ground shaking; and (4) earthquake-induced landslides, which are mass movements caused by loose, thin soil on steep slopes, especially when shaken during an earthquake.

A seismograph is a device that detects, amplifies, and records both ground motion and earthquakes. It operates on the straightforward idea of inertia (Sharma & Dhyani, 2020). As vibrations are essential components of fundamental

phenomena like light and sound, they play a significant role in the natural world. Many people are interested in ground vibrations since they result from sources other than naturally occurring earthquakes (Mayrhofer, 2019). P waves, also known as the primary wave, are the quickest seismic waves. P waves travel at a speed of between 5 and 8 kilometers per second (3.1 and 5 miles per second), and they move through rock in a manner similar to how sound waves move through air. This is why early seismologists named them P waves because they were the first waves to reach seismometers from some distant quakes. S waves moved just 60% as quickly as P waves. When an S wave penetrates a material, the area where it passes travels up and down or side to side at a speed of 3 to 4.8 kilometers per second (1.9 to 3 miles per second). S waves, a form of wave, are hence also known as transverse waves. Neither liquids or gasses can pass through this kind of wave (Perkins, 2018). Making something less intense, hazardous, painful, harsh, or destructive is known as mitigating it. The seismograph's function is to lessen the impacts of earthquakes that can result in harm to the economy, society, and culture (Merriam-Webster n.d.).

Seismometers can measure it internationally as well. In big earthquakes, the majority of the danger to buildings is caused by these waves, which are the cause of the ground shaking. Second, the shift in elastic energy that arises from the abrupt slip across the fault surface causes a

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permanent displacement of the Earth's surface (Tilloy et al., 2019). Mathematician and astronomer Zhang Heng created the first seismograph in China. Modern seismographs were developed in the 19th century. An electromagnetic seismograph was initially created by Luigi Palmieri in 1856 to capture the moment of the earthquake. A sensitive galvanometer is made possible by the generated electromagnetic field created by the coil and magnet's quivering action (Promita, 2021).

Huebsch (2019) stated that the practice of detecting motion dates back to the pre-1800s and has its origins in astronomy, which has a long history. Heinrich Hertz invented radar, the first motion-detecting technology. Hertz investigated the characteristics of waves and discovered that they could reflect off of things and travel at various speeds. A motion sensor, sometimes known as a motion detector, is a type of electronic device used to track and record movement (Jost, 2019). Samuel Bagno created the first motion detector in 1940. The Doppler effect and ultrasonic technology were invented by Bagno to detect motion in space. The development of technology around 1980 gave rise to infrared motion sensors (Coulombe, 2021). According to Jost (2019), motion is detected using this type of sensor, which gauges changes in the amount of sound or radiation reflected back into the receiver. Motion detectors are classified into two types: active and passive. When something disrupts or modifies the sensor's field, an electric pulse is sent to the embedded computer, which subsequently interacts with the mechanical part.

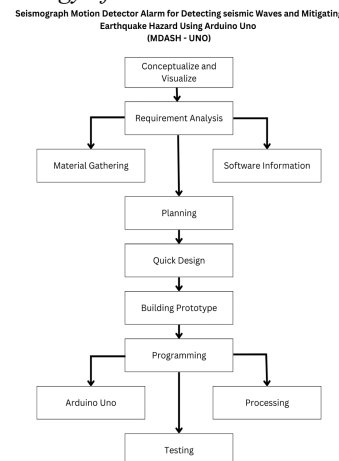
2. METHODOLOGY

2.1. Research Design

Experimental research design is the process used in this study. This type of research design is suitable for carrying out research that is being investigated in a controlled environment that can be tested by the hypothesis they make.

2.2 Agile Methodology

Figure 2.2
Agile Methodology of MDASH-UNO



2.3 Methods and Procedures

Conceptualize and Visualize

The initial title of the research study was "Horizontal Seismograph Motion Detector Alarm for Detecting Seismic Waves and Mitigating Earthquake Hazards". The conceptualization and visualization of the prototype will be covered in this section.

Requirement Analysis

The two sections of the requirement analysis are software information and material gathering. This stage involves obtaining and researching knowledge about the tools and software.

Material Gathering

Multiple studies that are comparable to the prototype were studied, and we came up with a list of requirements and things that are needed to execute this effectively. The things that are needed are: (1) an ADXL335 3 axis accelerometer used to detect if any ground shaking occurs; (2) an Arduino Uno Board that acts as the main component of the project; (3) a buzzer that is used as the signaling tool of the prototype; (4) an LED that will act as the visible indicator; (5) an LCD display used to show the intensity of the motion; (6) a

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breadboard that will serve as the temporary circuit board; and (7) wires that are used to connect all the electronic components

Software Information

Software applications, such as the Arduino IDE application and the Processing application, are studied. The Arduino IDE application is open-source software. Which is used to write and upload code to the Arduino IDE application and the Processing application, are studied. The Arduino IDE application is open-source software that is used to write and upload code to the Arduino. It contains a text editor for writing code, a message area, a text console, a toolbar with buttons for common functions, and a series of menus. Processing is a flexible software sketchbook and a language for learning how to code.

Planning

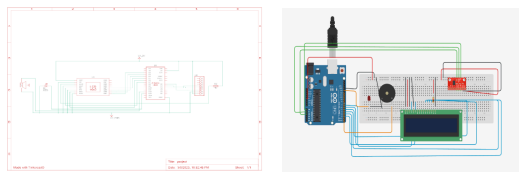
The planning part includes the process and brainstorming about the possible ways to put together the prototype from raw materials up to its final look and what tests will be done to meet the answer on the Statement of the Problem part.

Quick Design

This part presents the design of the prototype, from its wiring up to the final casing. As shown below, Figure 2.2

Figure 2.2

Electronic Materials gathered are put together



Building Prototype

The materials gathered during the material gathering time will be attached together following the quick design of the device. During these parts, buying, replacing, and adding electronic materials will be done. Connecting the male-to-male Dupont wire cable to each electronic material will be done, as well as creating the protective casing of the MDASH-UNO device.

Initial Casing

Sintra board serves as the initial casing's durable and lightweight material. Sintra board is a closed-cell polyvinyl chloride (PVC) board that has been moderately expanded and is both lightweight and rigid. The casing has dimensions of 12" by 6"

Testing Casing

An aluminum angle bar is used in the testing enclosure, which is specifically designed for the device's testing. Angle bars made of aluminum are perfect for a variety of manufacturing operations since they are lightweight and resistant to corrosion. Marine wood is adhered using a water-resistant glue because these woods have multiple layers of wood veneers. A piano hinge is a long, narrow hinge that can support greater weight than a typical hinge, while a blind rivet is a tubular fastener with a mandrel in the center. The testing case measures 25 by 35 centimeters.

Final Casing

Fiberglass serves as the final casing of the MDASH-UNO device, which has a measurement of 10" by 5".

Programming

This part is divided into two parts as the device uses two softwares. This is also expected to be a time-consuming part, as it is a trial-and-error process that will require debugging and checking. The two pieces of programming software that have to be used are the Arduino IDE and Processing Application.

3. RESULTS AND DISCUSSION

The MDASH-UNO device's motion detector and alarm are attached and work well, with the ADXL335 3-axis accelerometer detecting motion and triggering the alarm when it is detected. The attachment of motion detector alarms can help mitigate earthquake hazards by providing early warning to individuals and organizations. The Department of Science and Technology (DOST) MIRDC-ATL conducted the sine vibration (SV) testing, which was able to detect motion effectively using a motion sensor (ADXL335 3-axis accelerometer) to detect the frequency of an earthquake and send an alert through the buzzer's sound emission to warn people of potential ground shaking. This alert can give people time to take action, such as seeking shelter or stopping machinery, to reduce the risk of earthquake hazards. Therefore, the researchers conclude that the Seismograph

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Motion Detector Alarm for Detecting Seismic Waves and Mitigating Earthquake Hazard (MDASH-UNO) device is effective in terms of attaching the motion detector and alarm to help mitigate earthquake hazards, as it was tested using the Sine Vibration (SV) test conducted in the Department of Science and Technology (DOST) MIRDC-ATL laboratory, where the MDASH-UNO was able to detect the vibration made by one of the MIRDC-ATL device, and as it detected the vibration, MDASH-UNO was also able to send an alert through the buzzer's sound emission that could help mitigate future earthquake hazards by alerting people about the motion detected by the sensor.

The test that was conducted in the Department of Science and Technology (DOST) MIRDC-ATL laboratory to know how effective the MDASH-UNO is in detecting motion in terms of setting the parameters in Signal Plot Point, Drive Limit, Compression Rate, Bandwidth, Ramp Rate, Initial Drive, Sweep Type, Measurement Strategy, and Abort Sensitivity shows and describes the limits of how the MDASH-UNO can detect and stand strong motions. The sine vibration (SV) test was conducted three times for the X-axis, Y-axis, and Z-axis. The parameters that were selected for the MDASH-UNO for the x-axis are shown in Table 3.1

Table 3.1
Test Parameters of MDASH – UNO x,y, and z axis

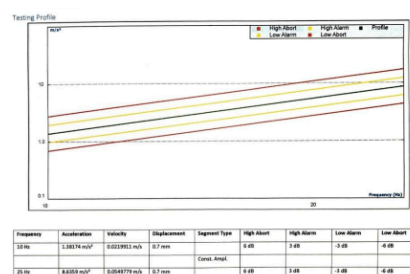
Signal Plot Points: 2048	Initial Drive: 0.005 Volts	Drive Limit: 2 Volt Pk.
Sweep Type: Logarithmic	Compression Rate: Fast	Measurement Strategy: Proportional Filter
Bandwidth: 25%	Abort Sensitivity: 0.50	Ramp Rate: Slow

Therefore, the researchers conclude that the Seismograph Motion Detector Alarm for Detecting Seismic Waves and Mitigating Earthquake Hazard (MDASH-UNO) device is effective in detecting motion in terms of setting the parameters in Signal Plot Points, Drive Limit, Compression Rate, Bandwidth, Ramp Rate, Initial Drive, Sweep Type,

Measurement Strategy, and Abort Sensitivity during the test conducted in the Department of Science and Technology (DOST) MIRDC-ATL laboratory.

As follows, the SV test conducted in the Department of Science and Technology (DOST) MIRDC-ATL laboratory used the set test parameters for each axis and sought the answer to the last question in the statement of the problem of the study, which is how much frequency can the MDASH-UNO device hold.

Figure 3.1
Testing Profile of MDASH-UNO x,y, and z axis



As shown in Figures 3.1, the frequency of the MDASH-UNO device can range from 10 Hz up to 25 Hz during testing in the Department of Science and Technology (DOST) MIRDC-ATL Laboratory. The 10 Hz to 25 Hz range refers to the frequency rate at which current changes direction every second. This rate is expressed in hertz (Hz), where 1 hertz equals 1 cycle per second. Therefore, the researchers conclude that the Seismograph Motion Detector Alarm for Detecting Seismic Waves and Mitigating Earthquake Hazard (MDASH-UNO) device is strong enough and effective in detecting motion, as MDASH-UNO was able to hold 10 Hz to 25 Hz during the test conducted in the Department of Science and Technology (DOST) MIRDC-ATL laboratory.

4. CONCLUSIONS

The Seismograph Motion Detector Alarm for Detecting Seismic Waves and Mitigating Earthquake Hazards (MDASH-UNO) device has significant effectiveness in detecting motion and mitigating earthquake hazards. The

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researchers conclude that the MDASH-UNO device is effective in terms of attaching the motion detector and alarm to help mitigate earthquake hazards and in detecting motion in terms of setting the parameters in Signal Plot Points, Drive Limit, Compression Rate, Bandwidth, Ramp Rate, Initial Drive, Sweep Type, Measurement Strategy, and Abort Sensitivity during the test conducted in the Department of Science and Technology (DOST) MIRDC-ATL laboratory and that the device is strong enough and effective in detecting motion, as MDASH-UNO was able to hold 10 Hz to 25 Hz during the test. MDASH-UNO was also able to send an alert through the buzzer's sound emission that could help mitigate future earthquake hazards by alerting people about the motion detected by the sensor. The device's advanced technology allows it to detect even the slightest vibrations, ensuring that no seismic activity or motion goes unnoticed. Furthermore, the MDASH-UNO's built-in alarm system is loud and attention-grabbing, ensuring that you will be alerted to any seismic activity even if you are occupied with other tasks.

Overall, the MDASH-UNO device is a highly effective tool for detecting motion and minimizing the risks associated with earthquakes. With the MDASH-UNO, you can rest assured that you will be alerted to any seismic activity in your area, allowing you to take the necessary precautions to protect yourself and your property.

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7. APPENDICES

Table 7.1
Costing of MDASH-UNO

Materials	Price/Cost
ADXL335 3 Axis Accelerometer	100
Arduino UNO Board (first arduino uno board)	200
Buzzer	5
LCD Display	100
LED	6
Breadboard	100
Dupont Wire	10
Arduino Uno Board	548
Total	1,069

Figure 7.1
Letter to Department of Science and Technology (DOST) - MIRDC

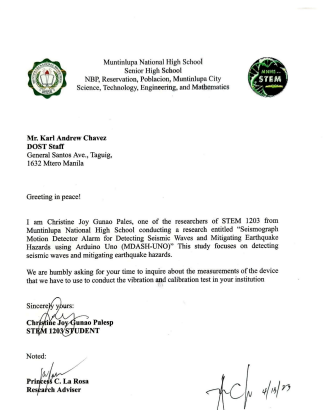


Figure 7.2
Test Results of MDASH - UNO x-axis

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PK Velocity (m/s)	0.09-0976	2 (Pa)	2.3%
Peak-Peak Displacement (mm)	0.7	53 Peak-Peak	3.4%
PK Range (gR)	33.693	1.020 (Pa)	9.9%

Run Schedule

2 Sweeps start from 100% between 1004 and 1004 at level 100.00%, total duration 00:01:02